

MANGALORE REFINERY AND PETROCHEMICALS LIMITED

STANDARD OPERATING PROCEDURE: SOP-OPS-014

Title: Post-Inspection Piping Integrity & Approval Protocol

1. PURPOSE & SCOPE

This standard defines statutory compliance actions following ultrasonic thickness inspection of hydrocarbon-bearing piping circuits in refinery process units.

2. CORROSION RATE CALCULATION FORMULA

The annual corrosion rate (CR) shall be computed as:

$$\text{Corrosion Rate (mm/year)} = (\text{Initial Nominal Thickness} - \text{Actual Measured Thickness}) / \text{Elapsed Service Years}$$

3. REMAINING SERVICE LIFE CRITERIA

$$\text{Remaining Life (Years)} = (\text{Actual Measured Thickness} - \text{Minimum Retirement Thickness}) / \text{Corrosion Rate}$$

Where:

- Nominal Crude Feed Piping Thickness = 5.0 mm
- Minimum Allowable Retirement Thickness = 3.0 mm

4. ACTION THRESHOLDS & ESCALATION

- If Remaining Life > 5.0 Years: Routine inspection cycle (36 months).
- If Remaining Life between 2.0 and 4.0 Years: Mandatory formal technical approval note, secondary ultrasonic scan within 90 days, and turnaround replacement package creation.
- If Remaining Life < 2.0 Years: Immediate de-rating or temporary composite wrap reinforcement.

5. SIGN-OFF AUTHORIZATION

Formal approval note must be generated and signed by Lead Inspection Engineer, Operations In-Charge, and Safety Department.